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Source: *The Review of Economics and Statistics*, May, 1977, Vol. 59, No. 2 (May, 1977), pp. 186-193

Published by: The MIT Press

Stable URL: <https://www.jstor.org/stable/1928815>

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USING THE *H*-INDEX OF CONCENTRATION WITH PUBLISHED DATA

Richard Schmalensee*

I. Introduction

IN theoretical discussions industrial organization specialists often indicate a preference for comprehensive or summary concentration indices over the more readily available concentration ratios. Such indices consider the entire size distribution of sellers in a market, and they give weight to both fewness of sellers and inequality of market shares. The most popular such measure is probably the *H* index of Hirschman (1945) and Herfindahl (1950). (See Hart (1975) for a discussion of alternatives and a list of references.) Let P_1 be the share of the largest firm in an industry, measured in terms of sales, employment, or whatever scale variable is considered most relevant, let P_2 be the share of the second largest firm, and so on, in a market with N sellers. Then *H* is defined by

$$H = \sum_{i=1}^N (P_i)^2. \quad (1)$$

If the P_i are all equal, it is easily seen that $H = 1/N$.

In spite of the theoretical appeal of *H* and other summary measures, most empirical work in industrial organization employs concentration ratios. Since exact values of summary indices are rarely available, some sort of surrogate must be employed. Further, most concentration measures have been observed to be highly correlated with one another. (See, for instance, Scherer, 1970, pp. 50–52, and Bailey and Boyle, 1971.) These high correlations seem generally to be taken to imply that the choice of a concentration measure can be made largely on the basis of convenience, and concentration ratios are convenient.

I do not find this convincing. An infinite number of alternative concentration measures

can be computed from typically available data. As long as correlations are not perfect, different measures will sometimes give different signals. Thus, in any particular study, it may matter which of the infinite set of available measures is selected, and the existing literature provides no guidance as to how this selection should best be made.

This study attempts to provide such guidance. The *H* index is treated as the ideal measure of concentration. Twelve more or less plausible surrogates for this index are defined in section II, which also describes the data used to compare them. Tests designed to weed out surrogates with low fidelity to *H* are described in section III, and test results are presented. Section IV summarizes these results and outlines their implications for empirical work.

II. Surrogates and Data

The sort of data commonly available for measuring an industry's concentration in terms of any scale variable can be transformed into the following:

s_i = total share of firms in rank i ; $i = 1, \dots, M$

N_i = number of firms in rank i ; $i = 1, \dots, M$

$a_i = s_i / N_i$ = average share of firms in rank i ;
 $i = 1, \dots, M$

N = total number of firms.

The first rank consists of the largest N_1 firms, the second of the next N_2 firms, and so on. In U.S. data, s_1 would be the four-firm concentration ratio, and $(s_1 + s_2)$ would be the eight-firm ratio. If one is given N and a set of concentration ratios, all these quantities are easily computed, and M is equal to one plus the number of ratios. Note that the sequence of a_i must be non-increasing.

We now derive a set of twelve surrogates for *H* that can be computed from this sort of information.¹ If size distributions of firms all

Received for publication December 8, 1975. Revision accepted for publication May 3, 1976.

*This article is a condensed and revised version of Schmalensee (1975), which was presented at the 1975 Winter Meeting of the Econometric Society in Dallas. I thank Clive Granger and John Hooper for help, with the usual caveat.

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¹In addition to these, $(1/N)$ was considered, but its performance on all the tests of section III was so terrible as to render the details uninteresting. The first eight measures presented below are identical or nearly identical to indices employed by Bailey and Boyle (1971).

belonged to some particular family, an elegant approach to such derivations would begin with efficient estimators of the parameters of each industry's distribution. As no such family seems to exist (see Silberman (1967)), however, it seems necessary to proceed in a less formal fashion.

Suppose that a_1 , the average share of the largest N_1 firms, is taken as the share of all firms. Substitution into (1) then yields the first surrogate:

$$CR4 = (1/a_1)(a_1)^2 = a_1. \quad (2)$$

In our data, this is one-fourth of the four-firm concentration ratio. Similarly, using $(s_1 + s_2)/(N_1 + N_2)$ as an estimate of all shares in (1) leads to $CR8$, which is one-eighth of the eight-firm concentration ratio here.

The minimum value of H that is consistent with the information assumed available is computed by assuming all firms in rank i have share a_i :

$$MIN = \sum_{i=1}^N N_i(a_i)^2. \quad (3)$$

A similar measure can be computed using only the first two s_i . Suppose all firms in rank 1 have shares a_1 and all other firms have shares a_2 . Then there must be $(1 - s_1)/a_2$ such firms, and the natural surrogate is

$$CR48 = N_1(a_1)^2 + [(1 - s_1)/a_2](a_2)^2 \\ = a_2 + s_1(a_1 - a_2). \quad (4)$$

In general, $CR4 \geq CR48 \geq CR8$, with equality if and only if $a_1 = a_2$.

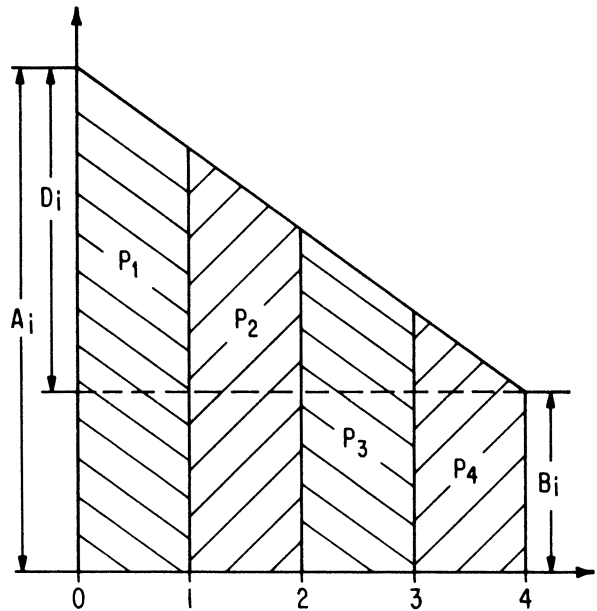
It is easy to see that the contribution to H from any rank is larger, the less equal the shares of the firms in that rank. On the assumption that all firms in rank i have shares a_i for $i \geq 2$, the contribution from rank one is maximized if it is assumed to consist of $(N_1 - 1)$ firms with shares a_2 and one large firm with share $[s_1 - (N_1 - 1)a_2]$. Assuming this jump in firm size, a little algebra leads to

$$MINJ = MIN + N_1(N_1 - 1)(a_1 - a_2)^2. \quad (5)$$

$CR48J$ is defined by adding the second term on the right to $CR48$.

The remaining surrogates allow firm sizes to vary linearly within ranks, as illustrated in figure 1 for some rank i with $N_i = 4$. The areas $P_1 - P_4$ are the estimated shares of the four

FIGURE 1.—LINEAR SHARE VARIATION WITHIN RANKS



firms, arrayed from largest to smallest. These must sum to s_i . Given a non-negative estimate of $D_i = A_i - B_i$, it can be shown directly that for any N_i , the sum of squares of the implied P 's is given by

$$h_i(D_i, a_i, N_i) \\ = N_i(a_i)^2 + (D_i)^2[(N_i)^2 - 1]/(12N_i). \quad (6)$$

Given estimates of the D_i , a surrogate for H can be computed by summing the h_i for all ranks.

The simplest way to employ (6) is to assume that the shares of the firms in the top rank decline linearly with $B_1 = a_2$, and to let $D_i = 0$ for $i \geq 2$. This assumption yields $D_1 = 2(a_1 - a_2)$, and, from (6), one obtains the surrogate

$$MINL = MIN + (a_1 - a_2)^2[(N_1)^2 - 1]/(3N_1). \quad (7)$$

$CR48L$ is defined by adding this same quantity to $CR48$.

To derive the surrogate LAV , consider ranks i and $i + 1$, and suppose the firm share curve of figure 1 extends to cover both ranks with the same slope, so that $A_{i+1} = B_i$ and $D_i/N_i = D_{i+1}/N_{i+1}$. These assumptions, along with the formulae for s_i and s_{i+1} , yield equations that can be solved for D_i and D_{i+1} . For $i = 1$ or $i = M$, only one estimate of D_i can be computed

in this fashion. For intermediate ranks, the average of the two estimates of D_i can be employed. LAV is then computed by substitution into (6) and summation. Note that the final set of estimated D_i need not be compatible with a declining firm share curve, since the constraints $A_i \leq B_{i-1}$, $i=2, \dots, M$, are not imposed.

The final three surrogates, LCH , LCM , and LCL , add these constraints. It is assumed that linear firm share curves describe within-rank variation in ranks $1, \dots, m$, where $m \leq M$. For $i=2, \dots, m$, we require $A_i = B_{i-1}$, so that the firm share curve is assumed piecewise linear in this region. If $B_m = Z$, a little algebra yields the estimates

$$\hat{D}_{m-i} = 2T_{m-i} - 2(-1)^i Z, \quad i=0, \dots, m-1, \quad (8)$$

where the T 's are defined recursively by $T_m = a_m$ and

$$T_{m-i} = (a_{m-i} - a_{m-i+1}) - T_{m-i+1}, \\ i=1, \dots, m-1.$$

We require the $\hat{D}_i$ to be non-negative and $Z \geq a_{m+1}$, where $a_{M+1} \equiv 0$. These conditions will be satisfied if and only if

$$Z_{\max} \equiv \min[T_m, T_{m-2}, \dots] \geq Z \\ \geq \max[a_{m+1}, -T_{m-1}, -T_{m-3}, \dots] \\ \equiv Z_{\min}. \quad (9)$$

The first step in computing these surrogates is to set $m = M$ and compute $Z_{\max}$ and $Z_{\min}$. If (9) cannot be satisfied, D_m is set to zero, m is reduced by one, and (9) is re-examined. Since $Z_{\max}$ is never less than $Z_{\min}$ when $m=1$, this process must terminate. When it does, LCM is computed using the $\hat{D}_i$ implied by (8) with $Z = (Z_{\max} + Z_{\min})/2$. If m is odd (even), LCH uses $Z = Z_{\min}$ ($Z_{\max}$) and LCL uses $Z = Z_{\max}$ ($Z_{\min}$). This gives LCH (LCL) the maximum (minimum) possible contribution from the first rank.

In order to evaluate empirically these (or any other) surrogates, one needs a sample of industries for which the true values of H are known and for which the surrogates can be computed. As far as I know, the only published data meeting these requirements are in table A.1 of Nelson (1963), which uses value of shipments as a scale variable and presents

information on a sample of 114 4-digit manufacturing industries for 1947 and 101 industries for 1954. In this study, we used all Nelson's industries for which the true value of H , the four-firm and eight-firm concentration ratios, and the total number of firms were available. This yielded samples of 92 industries for 1954 and 108 industries for 1947, which were used to study levels of concentration. A third sample was composed of the 80 industries present in both the 1947 and 1954 samples; it was used to study changes in concentration.

Nelson (1963, p. 35) notes that his selection procedure was biased toward large, highly concentrated industries. Both biases are present in our samples, but the second is somewhat attenuated because the industries dropped for lack of data were all highly concentrated.²

In the 1947 sample, the 20-firm concentration ratio was present for all but 9 industries, 7 of which had 20 or fewer firms. When this ratio was available, four ranks were used in computing the surrogates described above; otherwise M was 3. Similarly, this ratio was available and employed for all but 20 industries in 1954, 6 of which had 20 or fewer firms.³

To see how the relations between the true value of H and the various surrogates depend on the level of concentration, separate analyses were carried out for all three total samples (T) along with their concentrated (C) and unconcentrated (U) subsamples. The concentrated subsamples were composed of all industries with four-firm concentration ratios of 0.50 or greater. (The 1947 value was employed with the change sample.) This ratio was employed because, in contrast to H , it is generally directly observable.

III. Tests and Results

Taking H as ideal but unobservable, one's choice from among a set of surrogates for use in any particular task should depend on how

²See Nelson (1963), chapter 2 for the details of the process that led to the original samples. The detailed characteristics of the samples employed here are discussed in Schmalensee (1975), section III. Of the 15 industries we dropped, H was not given for 14.

³In Nelson's (1963) data, only the largest 50 firms were used to compute the "true" H . We similarly restricted computation of the surrogates to ensure comparability; see Schmalensee (1975), footnotes 7 and 14, for details. In the 1947 sample, 13 industries were such as to require LCH , LCM , and LCL to be computed with $m < M$; 12 of these

TABLE 1.—RANKS OF LEVELS

	<i>T</i>		<i>C</i>		<i>U</i>	
	54	47	54	47	54	47
CR4	—	—	—	—	—	—
CR8	—	—	—	—	—	—
MIN	—	—	—	—	—	—
MINJ	—	—	—	+	—	—
MINL	+	—	—	—	—	—
CR48	—	—	—	—	—	—
CR48J	—	—	+	+	—	—
CR48L	—	—	—	—	—	—
LAV	—	—	—	—	—	—
LCH	—	+	—	—	+	+
LCM	—	—	—	—	+	—
LCL	—	—	—	—	—	—
<i>N</i>	92	108	32	52	60	56

closely results obtained using each surrogate can be expected to resemble those *H* would have yielded. We now seek to inform such expectations by examining actual relative performance.

It is often not clear precisely how a concentration measure will be used at the time it is computed. It therefore seems sensible to seek robustness by avoiding surrogates that perform any of a set of plausible tasks noticeably worse than their competitors. Further, since our data are somewhat old, and relative performance of any task will be seen to be somewhat sample-specific, we follow a conservative approach and attempt to avoid surrogates likely to perform exceptionally badly, rather than attempting to isolate those likely to perform exceptionally well.

The most obvious use of a concentration measure is to rank industries at some point in time from most to least concentrated. To evaluate performance of this task, we computed Kendall's τ between all surrogates and the true *H* for the six samples indicated in table 1.⁴ Those surrogates with the largest τ 's are indicated with plus signs.

Using subscripts to indicate particular surrogates, consider for any sample the matrix *Z*, with typical element

$$Z_{ki} = (\tau_k - \tau_i) / [\hat{V}(\tau_k) + \hat{V}(\tau_i)]^{1/2},$$

$$i, k = 1, \dots, 12, \quad (10)$$

had four-firm concentration ratios below the sample median. There were 18 such industries in the 1954 sample; 16 had four-firm ratios below the median.

⁴A great deal more comparative performance data relating to tables 1–4 are presented in Schmalensee (1975), section IV. Kendall's τ was chosen as the nonparametric association measure because of its comparatively simple sampling distribution; see Kendall (1962), chapter 5.

where $\hat{V}$ is the unbiased estimator of sampling variance given by Kendall (1962, p. 88). On the null hypothesis that the population values of τ_i and τ_k are equal (and, approximately, equal to the root-mean-square (RMS) of the two sample values), Z_{ki} has approximately the standard normal distribution for $i \neq k$. A small value of Z_{ki} can be taken to indicate that the sample performance difference between surrogates *i* and *k* is inconsequential. From standard tables, if *i* and *k* were chosen at random and all population τ 's were equal, the probability of observing $|Z_{ki}| > 1.282$ would be about $(11/12)$ (.20) = .183, since $Z_{ii} \equiv 0$. In order to identify instances of noticeably bad relative performance, we focus on the column of *Z* corresponding to the largest observed τ . Minus signs in table 1 indicate that the corresponding elements of that column exceed 1.282. Such elements will be treated as indicating bad relative performance.

Since the column of *Z* considered was chosen by looking at the sample, the sampling distribution of the elements used is complicated to the point of intractability. (This problem arises in all this section's tests.) In particular, it seems unlikely that the true probability of obtaining an indication of bad relative performance when the null hypothesis is correct is exactly 0.183. It may be close, however, but this is only conjecture. On the other hand, our approach does provide a metric on performance that serves to render comparable results obtained in different samples, for the purpose of shortening the list of surrogates. Since our purpose is to shorten this list, a critical value much larger than 1.282 would be less useful.

Concentration measures might also be computed in order to rank *changes* in concentration over time. Here, however, a new problem arises: monotonic transformations of concentration measures affect results. Suppose industry A changes from five equal-sized firms to four, while industry B changes from ten to eight. Using *H*, the concentration increase in A is greater. Using the numbers-equivalent (see Adelman, 1969), $NE = 1/H$, however, the increase in B is greater. A third approach (suggested by Scherer, 1970, p. 52) is to use the logarithmic measure *LH*, equal to the natural logarithm of *H*. With this transformation, both industries show the same increase in concentra-

tion. It is not obvious which of these transformations is most informative here, and others might be considered as well. (Hirschman, 1945, for instance, uses $\sqrt{H}$.)

Given our concern for robustness, it seems reasonable to require that any defensible surrogate do an acceptable job of ranking changes in H , that its reciprocal do an acceptable job of ranking changes in NE , and that its logarithm do an acceptable job of ranking changes in LH . Using the 80 industries present in both 1947 and 1954, relative ranking abilities can be compared using exactly the procedures employed to construct table 1. The main results of these comparisons are presented as table 2.

TABLE 2.—RANKS OF CHANGES

	ΔH			ΔNE			ΔLH		
	T	C	U	T	C	U	T	C	U
CR4		+		—		—			—
CR8	—						—		
MIN				+	+	+			
MINJ				—		—			—
MINL					+	+			
CR48				—		—		+	—
CR48J				—		—			
CR48L				—		—			—
LAV									
LCH						+			
LCM						+			
LCL	+		+						+
N	80	37	43	80	37	43	80	37	43

Measures of the level of concentration are often employed as independent variables in cross-section regression analysis. Suppose that for a sample of N industries, one has hypothesized a model of the form

$$Y_i = C_i\beta_1 + X_i'\beta_2 + \epsilon_i, \quad i = 1, \dots, N, \quad (11)$$

where Y is some dependent variable of interest, C is the true (unobservable) concentration measure, X is a column vector of other independent variables, ϵ is a white noise error, and β_1 (a scalar) and β_2 (a vector) are to be estimated. Given a set of surrogates, $D^1, \dots, D^M$, which might be used as proxies for C , it is natural to further hypothesize a set of M linear stochastic relations between C and each of the D^j :

$$C_i = \alpha_j + \theta_j D_i^j + u_{ij},$$

$$i = 1, \dots, N; \quad j = 1, \dots, M, \quad (12)$$

where α_j and θ_j are surrogate-specific constants, and u_{ij} is white noise with variance σ_j^2 . Following McCallum (1972) and Wickens (1972), one might seek to select that proxy which would produce the smallest asymptotic bias in the estimated coefficients in (11). It is straightforward to show that the relative asymptotic biases in estimates of $(\theta_j\beta_1)$ and β_2 caused by using D^j in place of C in least squares estimation of (11) depends only on the β 's, correlations among C and the X 's, and on the squared correlation coefficient between C and D^j . (A proof is provided in the appendix to Schmalensee (1975).) Asymptotic biases are minimized by choosing that proxy variable with the smallest standard error in (12). This performance measure is also sensible in other situations in which surrogates are to be treated as cardinally valid, as would be the case if a surrogate were used to rank within-period differences between concentrations in pairs of industries.

Since the exact transformation of H that should appear in an equation like (11) is

TABLE 3.—PREDICTIONS OF LEVELS

	H						NE						LH					
	T		C		U		T		C		U		T		C		U	
	54	47	54	47	54	47	54	47	54	47	54	47	54	47	54	47	54	47
CR4	—	—			+	—	—	—			—	—			—			—
CR8	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
MIN		—		—					—	—	—	—		—	—	—		—
MINJ			+	+		—	—	—	+	+	—	—	—	—	+	+	—	—
MINL							+				+		+					
CR48	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
CR48J					—	—	—	—			—	—	—	—			—	—
CR48L					—	—	—	—			—	—	—	—			—	—
LAV	—		—		—	—	—	—	—		—	—	—	—	—	—	—	—
LCH	+	+				+	+				+		+				+	+
LCM																		
LCL																		
N	92	108	32	52	60	56	92	108	32	52	60	56	92	108	32	52	60	56

TABLE 4.—PREDICTIONS OF CHANGES

	ΔH			ΔNE			ΔLH		
	<i>T</i>	<i>C</i>	<i>U</i>	<i>T</i>	<i>C</i>	<i>U</i>	<i>T</i>	<i>C</i>	<i>U</i>
CR4	—	—	—	—	+	—	—	+	—
CR8	—	—	—	—	—	—	—	—	—
MIN									
MINJ			—	—		—	—		—
MINL	+	+					+		
CR48			—	—		—			—
CR48J			—	—		—			—
CR48L			—	—		—			—
LAV				—			—	—	
LCH			+	+		+			+
LCM									
LCL				—		—			
<i>N</i>	80	37	43	80	37	43	80	37	43

generally unclear, we must again be concerned with robustness. Regressions of the form of (12) were run for *H*, *NE*, and *LH* (using the obvious transforms of the surrogates) for the 1947 and 1954 samples and subsamples. The regressions with the lowest standard errors are indicated with plus signs in table 3. *F*-tests for the equality of error variances between the best regression in each sample and each of the others were performed. Using the upper 10% critical value of the relevant *F* distribution as the cutoff point for bad relative performance leads to the same probability of a Type I error as in tables 1 and 2. Instances of bad relative performance detected by these tests are again indicated with minus signs.

Precisely the same procedures were applied to the 80 observations of changes in concentration between 1947 and 1954, using the same three functional forms. The results are presented in table 4. It should be noted that the maintained hypothesis here is that of a linear stochastic relation between *changes* in *H* (or *NE*, or *LH*) and *changes* in each surrogate (or its reciprocal, or its logarithm).

IV. Summary and Implications

Observed instances of bad relative performance, as defined above, are summarized in table 5, which considers all tests (columns of tables 1–4) together and in various groupings.⁵ The quantity *Z* would be (approximately) a standard normal deviate on the null hypothesis that the probability of any surrogate performing relatively badly on any test is in fact 0.183.

⁵The results of table 1 are reflected in each of columns 3–5 in this table, since those results are invariant under monotone transformations of the concentration index.

All but two of these are highly significant, suggesting that significant performance differences do exist among these 12 measures. Asterisks indicate cases in which the probability of obtaining at least the indicated incidence of observed bad performance is less than 0.10 under the same null hypothesis. The frequency and systematic pattern of asterisks again suggests that some surrogates really are worse than others, contrary to conventional wisdom. But it should be noted that this evidence can only be suggestive, since the true probability of observing bad relative performance when all surrogates are equally good is unknown, and since the tests cannot really be considered independent.

Several general implications of table 5 deserve mention. First, a comparison of the totals and *Z*-values for the first two columns indicates that, as one would expect, the parametric tests of tables 3 and 4 were more powerful than the nonparametric procedures of tables 1 and 2. Second, I had initially expected that tests based on changes in concentration would be noticeably more powerful, relative to sample size, than those based on levels. A comparison of the sixth and seventh columns of table 5 fails to confirm this expectation. Third, the eighth and ninth columns suggest that there are more dramatic differences in surrogate performance in unconcentrated than in concentrated industries. While some of the observed differences are no doubt attributable to the smaller sizes of the concentrated subsamples, this impression is reinforced by consideration of columns 3–5. Both *NE* and *LH* tend to exhibit larger variation in unconcentrated than in concentrated industries, so that more weight is attached to performance in the former in tests involving these measures than in tests based on *H*. Since highly concentrated industries are overrepresented in the Nelson (1963) data, it appears that analysis of his entire 1954 or 1947 sample tends to understate the extent of differences among surrogate measures.

A glance at the first two rows of table 5 indicates that the eight-firm concentration ratio is quite inferior to available alternatives as a surrogate for *H*. The four-firm ratio is marginally better, but these findings (and our conservative approach) suggest that it can be

TABLE 5.—INSTANCES OF BAD RELATIVE PERFORMANCE PER COLUMN, TABLES 1–4

	1&2	/	3&4	H	/	NE	/	LH	1&3	/	2&4	C	/	U	Total
CR4	.20		.48*	.28		.44*		.17	.33*		.44*	—		.57*	.38*
CR8	.53*		1.00*	.89*		.83*		.89*	1.00*		.61*	.79*		.79*	.83*
MIN	—		.26	.11		.11		.17	.29		—	.36		—	.17
MINJ	.53*		.48*	.33		.67*		.61*	.50*		.50*	—		.86*	.50*
MINL	—		—	—		—		—	—		—	—		—	—
CR48	.60*		.85*	.67*		.83*		.83*	.96*		.50*	.50*		.93*	.76*
CR48J	.27		.53*	.17		.50*		.44*	.42*		.44*	—		.71*	.43*
CR48L	.33		.52*	.17		.50*		.50*	.42*		.50*	—		.79*	.45*
LAV	.07		.74*	.33		.39*		.56*	.67*		.28	.43*		.64*	.50*
LCH	—		—	—		—		—	—		—	—		—	—
LCM	—		—	—		—		—	—		—	—		—	—
LCL	—		.07	—		.11		—	—		.11	—		.07	.05
Total	2.53		4.93	2.94		4.39		4.17	4.58		3.39	2.07		5.36	4.07
Columns	15		27	18		18		18	24		18	14		14	42
Z	0.87		10.50	2.27		6.84		6.14	8.63		3.68	−0.46		8.71	8.99

* See text for explanation.

used with confidence (by one who takes *H* as the ideal measure) only in samples limited to highly concentrated industries. I had expected *CR48*, *CR48J*, or *CR48L* to outperform both these quantities. Instead, intermediate levels of performance were more usually attained, and table 5 indicates that these measures cannot be recommended for general use either. While they are simpler to compute than the remaining surrogates, the latter (as computed here) use at most four pieces of information, while *CR48*, *CR48J*, and *CR48L* employ two numbers each, so that differences in computational effort are slight.

Of the remaining surrogates, all of which consider all ranks, *LAV* and *MINJ* are clearly the worst by our standards. *LAV* is at best an interesting failure. *MINJ* performs relatively well in concentrated samples, but so do others that are no more difficult to compute.

Dropping *LAV* and *MINJ*, we are left with *MIN*, *MINL*, *LCH*, *LCM*, and *LCL*. *MIN* and *LCL* perform badly on occasion, though the frequency of such occurrences casts little doubt on the null hypothesis that all five are really equally good. Still, the conservative thing to do is to drop these two measures from further consideration and to limit our attention to the three surrogates that *never* exhibit significantly bad relative performance: *MINL*, *LCH*, and *LCM*.⁶

⁶ It is interesting to note that all three of these assume that firm shares decline linearly in the first rank. This supports the conjecture of Bailey and Boyle (1971, p. 704) that "in many instances, the linear firm-size assumption may be the most accurate [of any of those used here]." On the other hand, these authors go on to recommend use of either (a measure very similar to) *CR48* or *CR4*, while our analysis suggests that better surrogates are available at modest additional cost.

It is difficult to rank these three with any confidence, since (by our selection procedure) no performance differences between any two were ever significant at even the 20% level on a two-tailed test. *LCH* and *LCM*, in particular, are very highly correlated in all samples. Nontrivial performance differences between them occur in only 3 of the 42 columns of tables 1–4, and *LCH* performs better in all 3 cases. (These are column 6 in table 1 and columns 7 and 9 in table 4.) Further, *LCM* almost exhibits bad relative performance in column 7 of table 4, while neither *LCH* nor *MINL* ever comes at all close to doing so. There is apparently little to be lost by dropping *LCM*.

In the 14 tests involving unconcentrated subsamples, *LCH* outperformed *MINL* 12 times. If the probability of either being better were 0.50 and if the tests were independent, the probability of either performing better 12 or more times would be less than 0.01. However, the same test applied to each of the other column groupings shown in table 5 indicated totally insignificant performance differences. Further, the relative performances of the two differed noticeably on only 7 occasions, and *MINL* was the better measure on 3 of these. Since the overall performances of *LCH* and *MINL* are thus virtually indistinguishable, and since it is somewhat simpler to program the computation of *MINL*, that surrogate is probably the more attractive for most applications. If one is working mainly with unconcentrated industries, however, it seems that *LCH* may be likely to do a slightly better job.

It should be stressed that we have taken a conservative approach to summarizing per-

formance comparisons, worrying entirely about minimizing the risk of employing a surrogate that will perform badly. Others might feel that more stress should be placed on the likelihood of especially good performance. In any case, the analysis of this paper supports the recommendation that *MINL* and *LCH* be used as concentration measures by those who consider *H* an ideal index. We have found the eight-firm concentration ratio to be inferior to the four-firm ratio as a surrogate for *H*. Our results also indicate that the four-firm ratio can be used with any confidence only when dealing exclusively with highly concentrated industries.

REFERENCES

- Adelman, M. A., "Comment on the *H*-Concentration as a Numbers Equivalent," this REVIEW 51 (Feb. 1969), 99-101.
- Bailey, D., and S. E. Boyle, "The Optimal Measure of Concentration," *Journal of the American Statistical Association* 66 (Dec. 1971), 702-706.
- Hart, P. E., "Moment Distributions in Economics: An Exposition," *Journal of the Royal Statistical Society, Series A* 138 (1975), 423-434.
- Herfindahl, O. C., *Concentration in the Steel Industry*, unpublished Ph.D. dissertation, Columbia University, 1950.
- Hirschman, A. O., *National Power and the Structure of Foreign Trade* (Berkeley and Los Angeles: University of California Press, 1945).
- Kendall, M. G., *Rank Correlation Methods*, 3rd edition (New York: Hafner Publishing Company, 1962).
- McCallum, B. T., "Relative Asymptotic Bias From Errors of Omission and Measurement," *Econometrica* 40 (July 1972), 757-758.
- Nelson, R. L., *Concentration in the Manufacturing Industries of the United States* (New Haven: Yale University Press, 1963).
- Scherer, F. M., *Industrial Market Structure and Economic Performance* (Chicago: Rand McNally & Company, 1970).
- Schmalensee, R., "Using Good Concentration Measures in Spite of Bad Data," Discussion Paper 75-20, Department of Economics, University of California, San Diego, 1975.
- Silberman, I. H., "On Lognormality as a Summary Measure of Concentration," *American Economic Review* 57 (Sept. 1967), 807-831.
- Wickens, M. R., "A Note on the Use of Proxy Variables," *Econometrica* 40 (July 1972), 759-762.